Student Seminar - IDP Presentation 3

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Golden Dataset Validations

- Can be used to improve prompts/descriptions
- Main idea is to compare extractions to ground truth (golden dataset entries)

Extraction Evaluations

- Works without ground truth (other than paper content)
- Uses a mix of deterministic and LLM-supported steps

Evaluation

Extractions are evaluated in three aspects (per field):

Groundness: Is there evidence for extracted entries?
Faithfulness: Does the evidence actually match the entry?
Completeness: How many of the fields could be captured?

evidence_span: dict(field, evidence)

“Verbatim text sequences (1 per field - at least 5 words) from the source text grounding each extracted property.”

Groundness

Do a deterministic check in fulltext to see if evidence actually comes from the text and is not an hallucination

1) Direct match
2) Fuzzy search with a threshold (> 90%)

groundness_rate: $\frac{|\text{evidence(f) exists in fulltext}|}{|F_{\text{extracted}}|}$ for $f \in F_{\text{extracted}}$

Faithfulness

Use an LLM to compare the field value with the corresponding evidence to figure out if the value actually comes from the evidence. Only on grounded fields.

Example:
- value: 25° C / evidence: room temperature (is_faithfull: true)
- value: 10 pH / evidence: in acidic environment (is_faithfull: false)

faithfulness_rate: $\frac{|f \text{ exists in evidence(f)}|}{|F_{\text{grounded}}|} \text{ for } f \in F_{\text{grounded}}$

Completeness

How many of the fields are captured. **This doesn't distinguish between LLM couldn't retrieve the data or the data actually isn't mentioned in the paper.** Hence should be seen as a usefulness metric for downstream tasks.

completeness_rate: $\frac{|F_{\text{extracted}}|}{|F|}$

Example

sequence: NGIWY

evidence_span : {

"sequence": "NGIWY-amide (Asn-Gly-Ile-Trp-Tyr-NH2)"...

}

Groundness: Fulltext (no) / fuzzy search (Match score: 97.3% - yes)

Faithfulness: yes -> Reason: The evidence quote explicitly provides the peptide sequence NGIWY in the name NGIWY-amide.

Considerations

- Only some of the fields are supposed to have a separate evidence (non-verbatim fields like temp, pH, concentration etc.) Others are directly searched (e.g detection_methods)

- Some fields are conditional and not included in completeness check (e.g. conjugate_partner)

- Some fields are not considered due to being calculated from other fields (e.g length)

Potential Use Case

For downstream tasks, each aspect can have a meaning and extractions be classified such as:

Weak: Hallucination -> groundness_rate != 1.0

Ok: 0.8 > completeness_rate > 0.7

Strong: completeness_rate > 0.8

Other Ways

- **Checking the reasoning of the model**: It is tricky due to unstructured reasoning

- **Reiterating on the evaluations**: Might improve performance but also not as easy and requires careful implementation

Next Steps

- Improve xml parsing + add pdf parsing (to markdown)
- Run pipeline and get results (see what we have) or work on a different evaluation method